LAB2: WRITE A PYTHON PROGRAM FOR COMPUTING MEAN, MEDIAN , MODE , STANDARD DEVIATION AND PERCENTILE

#program to calculate the mean of an array using numpy.

```
import numpy as np
ar1=np.array([1,1,1,1,1,1,1,1,1])
mean=np.mean(ar1)
print("The original array is",ar1)
print(" The mean of the given array is:",mean)
```

#program to calculate the median of an array

```
#median=middle most value.
ar2=np.array([12,13,14,15,16,17,18,19,20])
med=np.median(ar2)
print("The original array is :",ar2)
print("The median value of the array is:",med)
```

#Program to find the mode

#MODE: The most repetitive value in an array.

```
import statistics as st
import numpy as np
ar3=np.array([10,20,20,30,40,50,60,10,10,100])
md=st.mode(ar3)
print("The original array is:", ar3)
print("\n")
print("The mode of the given array above is ",md)
```

Program to calculate the standard deviation of an array

#S.D: How far a value is deviated from the mean

Higher the S.D: higher is the deviation

#Lower the S.D: Lower the deviation

```
ar4=np.array([17,16,17,14,13,17])
sd=np.std(ar4)
m1=np.mean(ar4)
print("The original array is:",ar4)
print("\n")
print("The mean of the array is:",m1)
print("The calculated value of S.D is: ",sd)
```

Program to calculate the percentile

```
per=np.array([10,14,15,17,45,50,55,60,80,100,33,63,72])
#Find out the 75 percentile of the data
a=np.percentile(per,75)
print(a)
```

CLASS WORK:

TAKE AN INPUT FROM THE USER AND PERFORM THE FOLLOWING TASK:

step 1: Ask the user to input size of the array

step 2: Ask the user to input the values in the array

STEP 3: calculate the : Mean, Median, Mode, S.D, Percentile

STEP 4: Find the dimension of the array given by the user